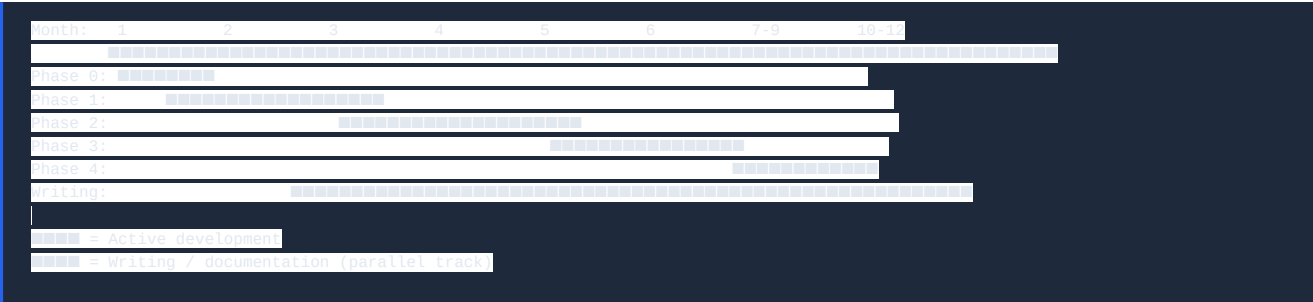


10 — Roadmap and Milestones

Interpretable Machine Learning for Discovering Hidden Regulatory Switches in Non-Coding DNA

Field	Value
Document ID	RM-001
Version	1.0
Status	Active
Created	2026-06-07
Related docs	Charter , Modeling Roadmap , Risk Register

Roadmap Overview



Month 1: Foundation

Objective

Establish data pipeline, environment, and first baseline model.

Milestones

#	Milestone	Target date	Success criterion
M1.1	Project setup complete	Week 1	Git repo initialized; directory structure created; requirements.txt defined; docs pack reviewed by team
M1.2	Literature survey complete	Week 2	≥ 20 papers read and annotated; annotated bibliography committed
M1.3	Data pipeline functional	Week 3	ENCODE cCREs downloaded for K562; sequences extracted from hg38; negative set generated; chromosome-holdout split defined
M1.4	First baseline model trained	Week 4	Logistic regression on k=4 k-mers; AUROC computed on test set; results logged

Deliverables

- ☐ Repository with directory structure and documentation
- ☐ Annotated bibliography
- ☐ Data pipeline: download → parse → extract → split
- ☐ EDA notebook with data visualizations
- ☐ First experiment log (`exp_001_logistic_kmer4`)
- ☐ Experiment registry initialized

Definition of Success at Month 1

A reproducible pipeline that goes from raw ENCODE annotations to a trained classifier with logged metrics. At least one model produces AUROC > 0.60 (proving the task is learnable).

Month 3: Interpretable Baseline Complete

Objective

Complete Phase 1 (classical ML) with full interpretability analysis and robust validation.

Milestones

#	Milestone	Target date	Success criterion
M3.1	Multi-model comparison	Month 2, Week 2	LogReg, RF, XGBoost trained and compared on same split; comparison table produced
M3.2	Feature importance analysis	Month 2, Week 3	Top-20 features extracted per model; cross-referenced with JASPAR; motif overlap table generated
M3.3	SHAP analysis complete	Month 2, Week 4	SHAP summary plot for XGBoost; SHAP force plots for 5 sample regions
M3.4	Negative set sensitivity analysis	Month 3, Week 1	Experiment E7 complete; AUROC stable ($\Delta < 0.05$) across ≥ 2 negative sampling strategies
M3.5	k-mer resolution analysis	Month 3, Week 1	Experiment E3 complete; optimal k value identified
M3.6	Feature ablation study	Month 3, Week 2	Experiment E2 complete; contribution of each feature group quantified
M3.7	Decision Gate 1→2 passed	Month 3, Week 3	AUROC ≥ 0.75 ; interpretability verified; all gate criteria met
M3.8	Phase 1 report written	Month 3, Week 4	Internal report summarizing baseline results, interpretability findings, data quality assessment

Deliverables

- ☐ ≥ 5 completed experiments with full logging
- ☐ Model comparison table (accuracy + interpretability)

- ☐ Motif cross-reference report
- ☐ SHAP analysis artifacts
- ☐ Feature ablation results
- ☐ Negative set sensitivity analysis
- ☐ Internal Phase 1 report
- ☐ Kaggle notebook reproducing key results
- ☐ Updated experiment registry

Definition of Success at Month 3

At least one classical model achieves AUROC ≥ 0.75 . Feature importance analysis reveals biologically plausible patterns ($\geq 30\%$ of top-20 features match known motifs). All results are reproducible. Decision Gate 1→2 is passed.

Month 6: Deep Learning and Cross-Cell-Type Analysis

Objective

Complete Phase 2 (CNN) and cross-cell-type experiments. Begin research writing.

Milestones

#	Milestone	Target date	Success criterion
M6.1	Shallow CNN trained	Month 4, Week 2	CNN achieves AUROC ≥ 0.80 on test set; training completes on Kaggle GPU
M6.2	CNN filter analysis	Month 4, Week 4	First-layer filters visualized as sequence logos; ≥ 3 filters match JASPAR motifs
M6.3	Saliency / Integrated Gradients	Month 5, Week 2	Saliency maps generated for 100+ test regions; qualitative analysis documented
M6.4	Cross-cell-type experiment	Month 5, Week 4	Experiment E4 complete for K562 + GM12878; cell-type-specific motifs identified
M6.5	Element-type classification	Month 5, Week 4	Experiment E5 complete; enhancer vs. promoter classifier evaluated
M6.6	Accuracy vs. interpretability report	Month 6, Week 1	Trade-off analysis comparing all model families documented
M6.7	Decision Gate 2→3 passed	Month 6, Week 2	All gate criteria met; team decides whether to pursue pretrained embeddings
M6.8	Draft research manuscript (introduction + methods)	Month 6, Week 4	Introduction and methods sections drafted

Deliverables

- ☐ Trained CNN model with filter analysis
- ☐ Saliency map gallery (representative examples)
- ☐ Cross-cell-type comparison report
- ☐ Enhancer vs. promoter classification results

- ☐ Full model comparison matrix (classical + CNN)
- ☐ Accuracy-vs-interpretability analysis document
- ☐ Draft manuscript (intro + methods)
- ☐ Updated Kaggle notebooks

Definition of Success at Month 6

CNN achieves AUROC ≥ 0.80 . Cross-cell-type analysis reveals cell-type-specific regulatory features. At least one novel (non-JASPAR) candidate motif pattern is identified by the CNN. Research manuscript is in progress.

Month 12 Vision: Research-Ready Platform

Objective

Complete remaining phases (as time allows), submit or post publication, and ensure the project is documented for long-term continuity.

Milestones (Months 7–12)

#	Milestone	Target window	Success criterion
M12.1	Pretrained embedding experiments (Phase 3)	Months 7–9	Embeddings extracted from ≥ 1 pretrained model; comparison to baselines documented
M12.2	Hybrid model (k-mer + embedding)	Month 8–9	Hybrid classifier evaluated; SHAP on hybrid features
M12.3	Full results manuscript	Month 9–10	Complete draft including results, discussion, limitations
M12.4	Model card(s)	Month 10	Model card for best model documenting performance, limitations, intended use
M12.5	Public code release	Month 10	Cleaned repository with README, license, installation instructions
M12.6	Publication submission	Month 11	Submitted to target venue (workshop or journal)
M12.7	Demo or visualization tool (optional)	Month 12	Gradio app or static HTML showing interactive regulatory predictions
M12.8	Project handoff documentation	Month 12	Updated all docs; contributor onboarding verified with a new team member

Deliverables (Months 7–12)

- ☐ Pretrained embedding comparison report
- ☐ Complete research manuscript
- ☐ Model card(s)
- ☐ Public repository release
- ☐ Submitted paper or preprint
- ☐ (Optional) Interactive demo
- ☐ Full project documentation update

Definition of Success at Month 12

The project has produced: 1. A published or submitted research paper on interpretable regulatory prediction. 2. A reproducible codebase usable by other researchers. 3. Documented evidence of regulatory switch-like sequence patterns. 4. A platform that can be extended by future contributors.

What "Success" Means at Each Stage

Stage	Success definition	Minimum acceptable outcome
Month 1	Pipeline works; first model trained	Data loads correctly; any AUROC > 0.5
Month 3	Interpretable baseline with motif validation	AUROC ≥ 0.75; ≥ 3 known motifs in top-20 features
Month 6	CNN + cross-cell-type analysis	AUROC ≥ 0.80; cell-type-specific patterns documented
Month 12	Publication + reproducible platform	Paper submitted; code public; handoff-ready

Minimum Viable Research Contribution (MVRC)

Even if the project stops at Month 3, the following constitutes a valid research contribution:

- 1. Reproducible pipeline for ENCODE-based regulatory classification.
- 2. Multi-model comparison showing accuracy-interpretability trade-offs.
- 3. Feature importance analysis connecting k-mer/motif features to known biology.
- 4. Honest documentation of what works and what doesn't.

This can be published as a workshop paper or technical report.

Milestone Dependency Graph



Progress Tracking

This section should be updated as milestones are completed.

Milestone	Status	Date completed	Notes
M1.1	■ Not started	—	—
M1.2	■ Not started	—	—
M1.3	■ Not started	—	—
M1.4	■ Not started	—	—
M3.1	■ Not started	—	—
M3.2	■ Not started	—	—
M3.3	■ Not started	—	—
M3.4	■ Not started	—	—
M3.5	■ Not started	—	—
M3.6	■ Not started	—	—
M3.7	■ Not started	—	—
M3.8	■ Not started	—	—
M6.1	■ Not started	—	—
M6.2	■ Not started	—	—
M6.3	■ Not started	—	—
M6.4	■ Not started	—	—
M6.5	■ Not started	—	—
M6.6	■ Not started	—	—
M6.7	■ Not started	—	—
M6.8	■ Not started	—	—
M12.1	■ Not started	—	—
M12.2	■ Not started	—	—
M12.3	■ Not started	—	—
M12.4	■ Not started	—	—
M12.5	■ Not started	—	—
M12.6	■ Not started	—	—
M12.7	■ Not started	—	—
M12.8	■ Not started	—	—

End of Document 10 — Roadmap and Milestones
Next: [11 — Glossary & Project Memory](#)